Example ([1, Example 6.16]). Let $(R, \mathfrak{m})$ be a local ring with $0 \neq \mathfrak{m}^2 = \mathfrak{m}$. Let $\tau^\sharp$ be the HRS tilt [2, Proposition I.2.1]:

$$(\mathrm{Mod}_R)^\sharp_{\geq 0} = \{X \in (\mathrm{Mod}_R)^{\mathrm{std}}_{\geq 0} : (\pi_0^{\mathrm{std}} X)\mathfrak{m} = \pi_0^{\mathrm{std}} X\}, \qquad (\mathrm{Mod}_R)^\sharp_{\leq 0} = \{X \in (\mathrm{Mod}_R)^{\mathrm{std}}_{\leq 1} : (\pi_1^{\mathrm{std}} X)\mathfrak{m} = 0\}.$$

It is equivalent to the standard t-structure, hence right complete, and compatible with filtered colimits because the two defining module classes are. If P is compact and $\tau^\sharp$-connective, then $\pi_0^{\mathrm{std}} P$ is finitely presented and $(\pi_0^{\mathrm{std}} P)\mathfrak{m} = \pi_0^{\mathrm{std}} P$. Nakayama gives $\pi_0^{\mathrm{std}} P = 0$, hence $P \in (\mathrm{Mod}_R)^{\mathrm{std}}_{\geq 1}$ and $\mathrm{Map}_R(P, \mathfrak{m}) \simeq *$. Thus compact connective objects do not generate $(\mathrm{Mod}_R)^\sharp_{\geq 0}$.

References

[1] L. Angeleri Hügel, *Silting Objects*, Bull. Lond. Math. Soc. **51** (2019), 658–690.

[2] D. Happel, I. Reiten, and S. O. Smalø, *Tilting in Abelian Categories and Quasitilted Algebras*, Mem. Amer. Math. Soc. **120** (1996), no. 575.